



In silico genome-wide analysis of the growth-regulating factor gene family and their expression profiling in *Vitis vinifera* under biotic stress

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Abstract

Growth regulatory factors (GRFs) are transcription factors that encode the proteins involved in plant growth and development. However, no comprehensive analysis of *Vitis vinifera* GRF genes has yet been conducted. In the current study, we performed a genome-wide analysis of the GRF gene family to explore the VvGRF gene's role in *Vitis vinifera*. We identified 30 VvGRF genes in the *Vitis vinifera* genome, localized over 20 chromosomes. Based on evolutionary analysis, 49 GRF genes (nine AtGRF, ten FvGRF, and 30 VvGRF) were clustered into six groups. Many cis-elements involved in light control, defense, and plant growth have been identified in the promoter region of VvGRF genes, and multiple miRNAs have been predicted to be involved in regulating VvGRF gene expression. Protein-protein interaction analysis showed that nine VvGRF proteins formed a complex protein interaction network. Furthermore, the gene expression analysis of VvGRF revealed that VvGRF-5 and VvGRF-6 were highly upregulated suggesting that these genes are involved in biotic responses. This study provides comprehensive insights into the functional characteristics and occurrence of the VvGRF gene family in *Vitis vinifera*, which may be applied in breeding programs to enhance the growth of *Vitis vinifera* varieties under stress and growth changes.

Keywords *Vitis vinifera* · Growth regulatory factor · Expression analysis · Phylogenetic analysis · Gene structure

Introduction

Vitis vinifera is a well-known grape species belonging to the Vitaceae family and is native to Western Asia and Southern Europe. The seeds and leaves of *Vitis vinifera* plants are used as herbal medicines, while the fruits are consumed as a food supplement [1]. About 50–75 percent of grapes are used to make wine, which is the most prevalent usage of grapes. In India, there are many varieties and species of *Vitis vinifera* [1]. *Vitis vinifera* is a good model for studying the development, color change, and disease resistance of perennial fruit plants. Agrobacterium-mediated method can be used to transform *Vitis vinifera* [2]. It has been proposed that the *Vitis vinifera* genome consists of three ancestral genomes and a more primitive evolutionary path. However, *Vitis vinifera* is considered a “paleohexaploid” plant [3]. Grape berries contain phenolic compounds, and their derivatives have antioxidant properties. Simple flavonoids such as anthocyanins and condensed proanthocyanidins (PAs, tannins) are dissolved into vacuoles or attached to polysaccharides in cell walls [4].

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Plant-specific transcription factors, also called growth regulatory factors, are essential for regulating plant growth and abiotic stress. In recent years, *GRF* genes have been widely studied in various species [5]. *GRF* has two conserved domains in its N-terminal region - QLQ (Gln, Leu, Gln) and WRC (Trp, Arg, Cys) [6]. *GRF*-interacting factors (*GIFs*) and their QLQ domains interact to make a functional *GRF*–*GIF* complex via the SYT N-terminal homology (SNH) domain. The WRC domain binds to cis-acting regions of downstream genes and regulates gene expression and localization [7]. Compared to the N-terminus of the *GRF* protein, the C-terminus is variable. Motifs like FFD (Phe, Asp), GGPL (Gly, Pro, Leu), and TQL (Thr, Gln, Leu) are often located near the C-terminal end of proteins.

The *GRF* gene family is crucial for the growth of plants. One study found that the *GRF* gene in rice mediates the regulation of gibberellin (GA)-induced stem development in deep water rice [8]. Since then on, numerous plant species have undergone genome-wide analysis of *GRF* gene families, including *Brassica napus* [9], *Triticum aestivum* [5], *Soybean* [10], *Arabidopsis thaliana* and *Fragaria vesca* [11]. Recent large-scale research on *Arabidopsis* and other crops have indicated that *GRF* genes play an essential role in mediating root, stem, fiber, and leaf development [12]. For example, rapeseed gene *BnGRF* regulates seed weight, oil content, and leaf size. Transgenic *Arabidopsis* plants carrying the *BnGRF-2a* gene had larger leaves (20% increase in leaf size), higher seed mass, and a 50% increase in oil yield [13]. In *Arabidopsis*, the plant growth factor *AtGRF-9* significantly suppressed leaf growth, while the *arf3* mutant decreased leaf size by about 15% when compared to the wild type [14]. In tomatoes, *GRF* genes regulate fruit development [15].

This study aimed to provide a comprehensive analysis of the *Vitis vinifera GRF* gene family. *Vitis vinifera* (*VvGRF*) genes were identified using the phytozome (DOE Joint Genome Institute). Further, we studied the conserved domains, exon-intron structure, phylogenetic analysis, protein-protein interaction, gene localization, and cis-regulatory element analysis of these genes. Finally, we analyzed the expression profiles of various organelles at various developmental stages compared to normal samples. Our findings provide the basis for future functional research on the *GRF* family in *Vitis vinifera* and future genetic modification efforts to enhance crop production under stress conditions.

Materials and methods

Sequence retrieval

The *GRF* gene sequences of *Arabidopsis thaliana* were retrieved from plantTFDB (<http://planttfdb.gao-lab.org/>).

Further, the *AtGRF* sequences of the model plant were BLAST against *Vitis vinifera* using the phytozome v13 (<https://phytozome-next.jgi.doe.gov/>).

Phylogenetic analysis and Collinearity analysis

Phylogenetic analysis was performed to study the evolutionary relationships between the *GRF* genes of *Vitis vinifera*, *Fragaria vesca*, and *Arabidopsis thaliana*. The complete gene coding sequence was downloaded from phytozome (<https://phytozome-next.jgi.doe.gov/>). Multiple sequence alignment was performed using the ClustalW server. Using *GRF* protein sequences and bootstrap analysis with 1000 repeats, the maximum likelihood technique of MEGA-11 software (<https://www.megasoftware.net/>) was utilized to generate phylogenetic trees with default parameters [16]. The phylogenetic tree was colored based on different clades using the iTol (<https://itol.embl.de/>) online database [17]. Whole genome sequencing and annotation files of *Arabidopsis*, *Vitis vinifera*, *Oryza sativa*, and *Fragaria vesca* were retrieved from Phytozome for Collinearity analysis. Collinearity analysis of different plant species was performed using the Multicollinearity Scanning Toolkit (MCScanX) plugin TBtool [8]. *Vitis vinifera* and three other plant species were shown to have homologous pairs using the Dual Synteny Plot in MCScanX [18].

Physio-chemical properties, conserved motif, domain analysis, and gene structure

Physicochemical properties such as amino acid length, molecular weight (MW), parental position, and theoretical isoelectric point (PI) were investigated using Phytozome v13 and the ExPASy ProtParam tool (<http://web.expasy.org/protparam/>) [8]. Multiple Expectation MEME (Maximization for Motif Elicitation) (<http://meme-suite.org/index.html>) was used for motif analysis. Ten motifs were selected for data analysis, while the other parameters were set to as defaults [19]. The online program Gene Structure Display Server (<https://gsds.gao-lab.org/>) was used to plot the *VvGRF* gene structures [20]. The NCBI Conserved Domain Search (<https://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi>) was utilized to download conserved domain hit data for domain analysis [21].

Cis-regulatory element (CRE) analysis

The 1500 bp upstream sequences of *VvGRF* were retrieved from Phytozome v13. CREs from promoter sequences were identified using the PlantCARE web database (<https://bioinformatics.psb.ugent.be/webtools/plantcare/html/>), and the results were analyzed using TBtools Simple BioSequence Viewer [22].

Sub-cellular localization of *VvGRF* genes in *Vitis vinifera*, chromosomal distribution, and gene duplication event analysis

The WoLF PSORT database (<http://wolfsort.hgc.jp/>) was used to analyze the subcellular localization of the *GRF* proteins. This database uses the *GRF* protein sequence file as input to generate a list of likely positions and their associated predicted scores. This analysis aimed to determine the proteins' location in the cell [20]. The *Vitis vinifera* gff3 file was obtained from Phytozome, and the chromosome length and location data of the 30 *GRF* genes were filtered through TBtools. MG2C v2.1 was used to describe chromosomal location [19]. Synonymous (Ks) and non-synonymous (Ka) rates of the *VvGRF* gene duplication pairs were calculated with TBtools [23].

Protein-protein interaction analysis

The STRING database (<https://string-db.org/>) was utilized to evaluate the protein-protein interactions of *GRFs*. The gene IDs were submitted as a query, selecting *Vitis vinifera* in the organism domain. Functional partners with the maximum combined confidence value (≥ 0.8) were chosen [24].

Expression data analysis of *Vitis vinifera*

To understand how biotic stress affects *GRFs* in *Vitis vinifera*, we downloaded transcript gene expression data from the NCBI GEO database (<https://www.ncbi.nlm.nih.gov/>) with accession number GSE175475 which contains expression data from embryogenic callus tissue of *Vitis vinifera* exposed to mock and live yeast treatment. To understand the expression mechanism of these genes, RNA-sequencing data was retrieved using the GEOquery R package and normalized into Log2. Further, using the DESeq2 R package, differential expression genes between the control and mock treatment were identified [20].

miRNA target analysis

The Plant MicroRNA Encyclopedia (<https://www.pmiren.com/>) retrieved miRNA sequences from *Vitis vinifera*. *Vitis vinifera* genome data substitutes of *VvGRFs* were submitted to psRNATarget (<https://www.zhaolab.org/psRNATarget/>) to predict miRNA targets by analyzing them against mature miRNA sequences [18]. The SRplot server was used to generate a visualized diagram of the miRNA-target interaction.

GO analysis

GO annotations were used to carry out GO enrichment studies to estimate the functional roles of *GRF* genes in

grapevines. The UniProt online database (<https://www.uniprot.org/>) contains detailed information on the biological processes, cellular components, and molecular functions of the *GRF* genes. ShinyGo software v.0741 (<http://bioinformatics.sdstate.edu/go74/>) was used for the GO enrichment analysis of the *VvGRF* gene sequences [20].

Results

Identification of the *GRF* gene family in *Vitis vinifera* and its physio-chemical properties

We identified 21 *Arabidopsis AtGRF* protein sequences retrieved from the Plant Transcription Factor Database (PTFD), which were used as *GRF* query sequences against a *Vitis vinifera* BLAST search. Thirty *Vitis vinifera GRF* genes have been identified in *Vitis vinifera*. The presence of QLQ and WRC domains in the *Vitis vinifera GRF* gene was confirmed using the SMART and Pfam sites. The 30 *Vitis vinifera GRF* genes were named *VvGRF*-1–*VvGRF*-30. Significant differences in amino acid sequence length were identified in *VvGRF*. There were 180 amino acids in the smallest *Vitis vinifera GRF* protein (*VvGRF*-26) and 1066 amino acids in the longest (*VvGRF*-19) (Fig. 1A). The range of the theoretical gravity value was -0.471 (*VvGRF*-25) to -1.171 (*VvGRF*-23) (Fig. 1B). The pH, or isoelectric point, at which a particular molecule is electrically neutral falls between 5.31 (*VvGRF*-23) and 10.36 (*VvGRF*-27) (Fig. 1B). The physiological properties of the *VvGRF* proteins are presented in Table 1.

Chromosomal distribution of *VvGRF* genes and gene duplication

The *GRF* genes are located throughout the chromosomes of *Vitis vinifera*. In *Vitis vinifera*, there are 20 chromosomes overall; 19 have been studied, and one is unknown. No *VvGRF* genes were located on chromosomes 1, 3, 4, 5, 6, 7, 10, 12, 13, 17, 19, and 20 (unknown), while other chromosomes contained 23 *VvGRF* gene members. Chromosome 15 had a maximum seven number of *GRF* members (Fig. 2). Chrs 9, 11, and 18 contain one *VvGRF* gene, while others have more than one *VvGRF* gene (Fig. 2).

Tandem and segmental duplications are significant mechanisms leading to genome expansion and greater complexity. Four duplicated *VvGRF* gene pairs were identified: *VvGRF*-24, *VvGRF*-23, *VvGRF*-18, *VvGRF*-16, *VvGRF*-30, *VvGRF*-29, *VvGRF*-13, and *VvGRF*-14 (Fig. 2).

Collinearity analysis

Gene family evolution is closely linked to the prevalence of segmental and tandem duplication. Gene duplication is vital

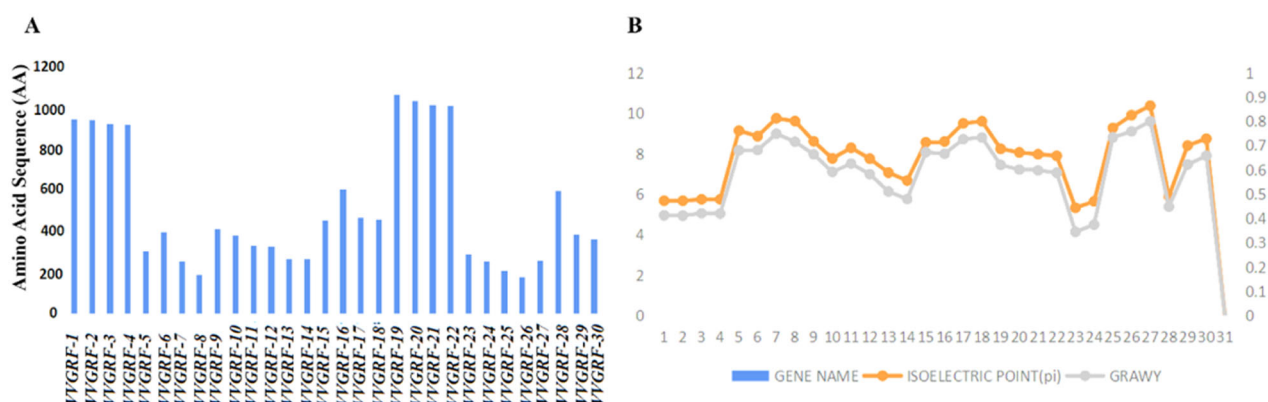


Fig. 1 A Amino acid length of *GRF* proteins B Isoelectric point and gravy values of *GRF* proteins

Table 1 Physiochemical properties of *VvGRF* genes

Transcription ID	Gene names	Chromosomes	Start location	End location	Molecular weight
VIT_214s0060g02420.2	VVGRF-1	11	2050800	2057311	107326.79
VIT_214s0060g02420.1	VVGRF-2	9	2050800	2057311	106867.23
VIT_214s0060g02420.4	VVGRF-3	16	2050800	2057311	105340.64
VIT_214s0060g02420.3	VVGRF-4	16	2050800	2057311	104881.08
VIT_208s0007g06690.1	VVGRF-5	16	20378674	20380185.00	33180.30
VIT_208s0007g06640.1	VVGRF-6	18	20328535	20330254	44084.97
VIT_208s0007g06640.3	VVGRF-7	UN	20328535	20330254	28205.06
VIT_208s0007g06640.2	VVGRF-8	16	20328535	20330254	21187.91
VIT_208s0007g03760.1	VVGRF-9	2	17725857	17728414	45222.55
VIT_216s0039g01450.3	VVGRF-10	2	976014	979605	42700.58
VIT_216s0039g01450.1	VVGRF-11	8	976014	979605	36838.78
VIT_216s0039g01450.2	VVGRF-12	15	976014	979605	36519.42
VIT_216s0039g01450.4	VVGRF-13	15	976014	979605	30113.99
VIT_216s0039g01450.5	VVGRF-14	16	976014	979605	29793.63
VIT_216s0098g01080.1	VVGRF-15	16	21346759	21348557	48871.75
VIT_218s0001g08650.1	VVGRF-16	14	7086699	7089964	65431.73
VIT_215s0048g01740.2	VVGRF-17	14	15941484	15945111	50874.37
VIT_215s0048g01740.1	VVGRF-18	14	15941484	15945111	50369.86
VIT_215s0046g00340.3	VVGRF-19	14	17348762	17355632	120278.53
VIT_215s0046g00340.1	VVGRF-20	UN	17348762	17355632	117028.53
VIT_215s0046g00340.4	VVGRF-21	8	17348762	17355632	114529.74
VIT_215s0046g00340.2	VVGRF-22	8	17348762	17355632	114583.91
VIT_215s0046g01860.1	VVGRF-23	8	18683834	18685159	32541.67
VIT_215s0046g01860.2	VVGRF-24	15	18683834	18685159	28927.86
VIT_208s0007g03760.2	VVGRF-25	15	4421650	4422503	23184.51
VIT_216s0039g01450.3	VVGRF-26	15	2331494	2332154	20364.93
VIT_216s0039g01450.1	VVGRF-27	15	11970314	11971168	30506.04
VIT_216s0039g01450.2	VVGRF-28	8	30596793	30599992	64118.15
VIT_216s0039g01450.4	VVGRF-29	15	1000358	1001843	44084.07
VIT_216s0039g01450.5	VVGRF-30	15	1135238	1136542	40942.46

for expanding the gene families on different chromosomes [25, 26]. In our study, all *VvGRF* genes located on the 20 chromosomes of *Vitis vinifera* were linked to all

chromosomes of *Arabidopsis thaliana* (Fig. 3A), *Fragaria vesca* (Fig. 3B), and *Oryza sativa* (Fig. 3C). Horizontal lines represent chromosomes, and the curves show the

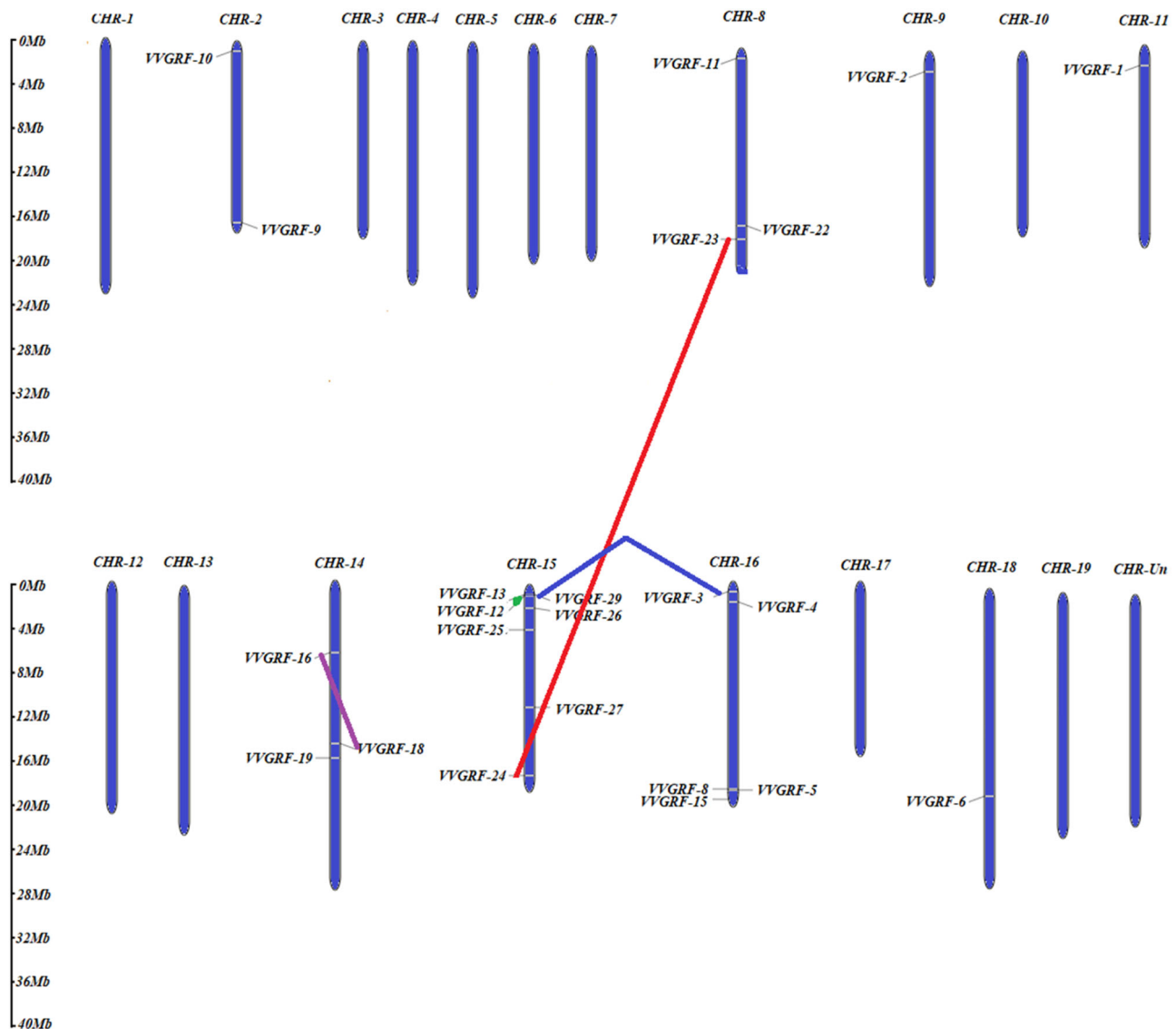


Fig. 2 Distribution of *VvGRF* genes across *Vitis vinifera* chromosomes. Parologue pairs are shown by red, blue, purple, and green lines

linkage between the genes. All three plant species exhibited homologous relationships with *Vitis vinifera*. Thus, these findings show that they diverged among several plant species after evolving from a common ancestor. Gray lines represent collinear blocks, and red lines highlight homologous gene pairs in the *GRF*.

Cis-regulatory elemental (CRE) analysis

TF binding sites that control transcription initiation or repression are cis-acting regulatory elements. Using CRE analysis, researchers could study the role of these genes in light sensitivity, hormone signaling pathways, and plant growth and development. We identified a total of 12 cis-regulatory elements, including AARE (abscisic acid responsiveness element), AIRE (anaerobic induction responsive elements), MeJARE (methyl jasmonate

responsiveness element), SARE (salicylic acid responsiveness element), and GRE (growth-regulating elements) (Fig. 4). Thus, *GRF* genes serve a variety of purposes during plant growth. Promotor analysis makes it simple to determine each gene's role. Specifically, elements critical for responding to abiotic stress were found, such as elements responsive to cold and dehydration, drought, low temperatures, defense, and stress. Based on these results, we hypothesize that *VvGRF* TFs tolerate various abiotic and biotic stimuli and play an important role in the growth of *Vitis vinifera*.

Gene structure

The diversification of *VvGRF* family gene structures indicates that newly evolved gene copies often developed new gene functions during the evolution of multiple gene

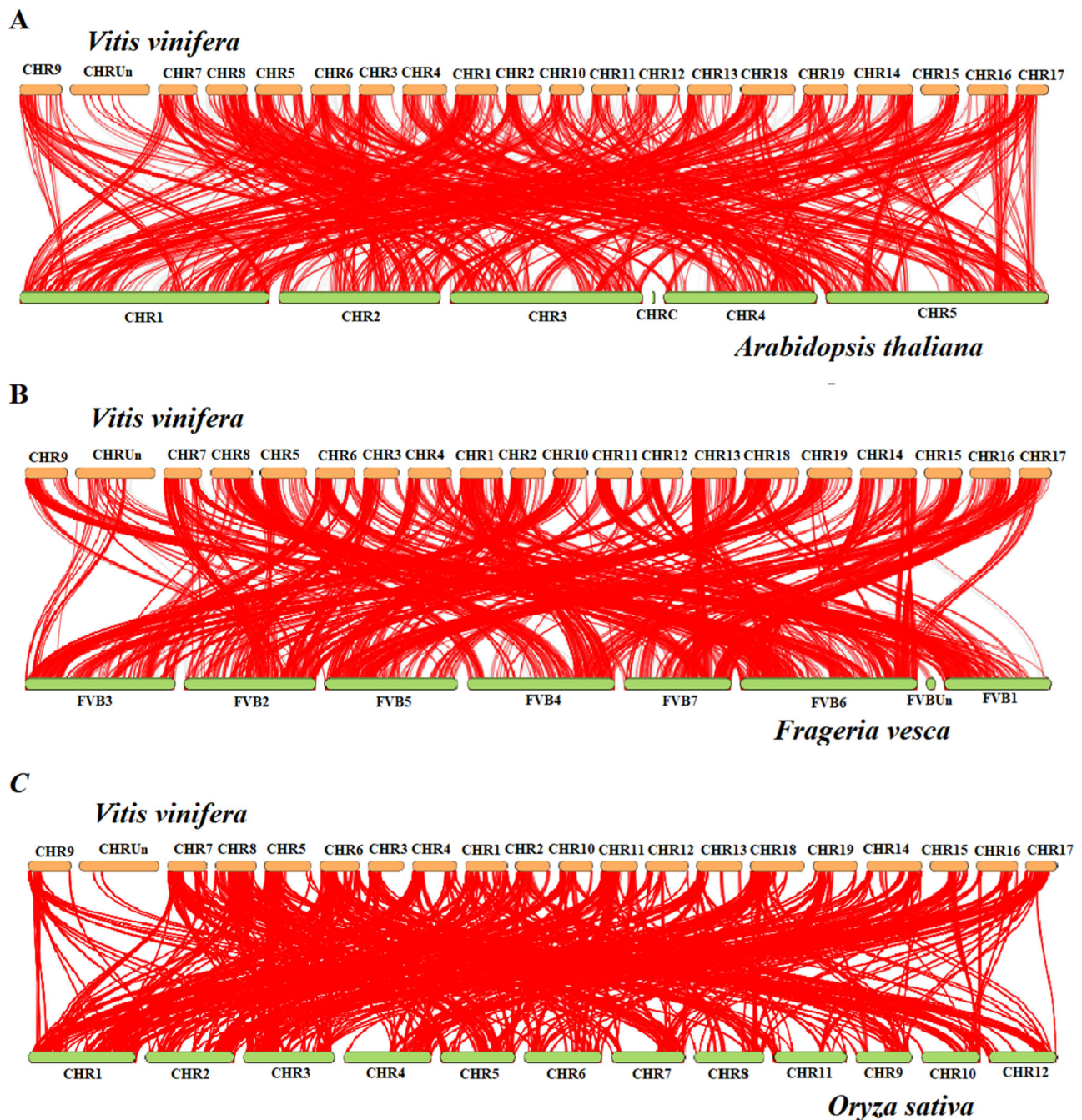


Fig. 3 GRF gene collinearity analysis of *Vitis vinifera* with **A** *Arabidopsis Thaliana*, **B** *Fragaria Vesca* and **C** *Oryza Sativa*

families. The exon/intron sequences of all *VvGRFs* were used to assess their structural organization to better understand the structural diversity of the different *VvGRF* genes. All the *VvGRF* genes consisted of introns and exons; the number of exons ranged from 1 to 13, and the number of introns ranges from 0–12 (Fig. 5). *VvGRF-17*, *VvGRF-19*, and *VvGRF-29* had the highest number of introns and exons; therefore, gene members with close relationships have similar gene structures. Exons and introns are indicated by yellow boxes and thin lines, respectively (Fig. 5).

Blue boxes represent the UTRs. The intron steps are numbered from 0 to 2.

Evolutionary analysis

A phylogenetic tree was constructed using *GRFs* from *Arabidopsis* (9 members), *Fragaria vesca* (10 members), and *Vitis vinifera* (30 members) to elucidate evolutionary relationships. The tree is divided into six subgroups (I–VI), represented by different colors. There were 13 members in

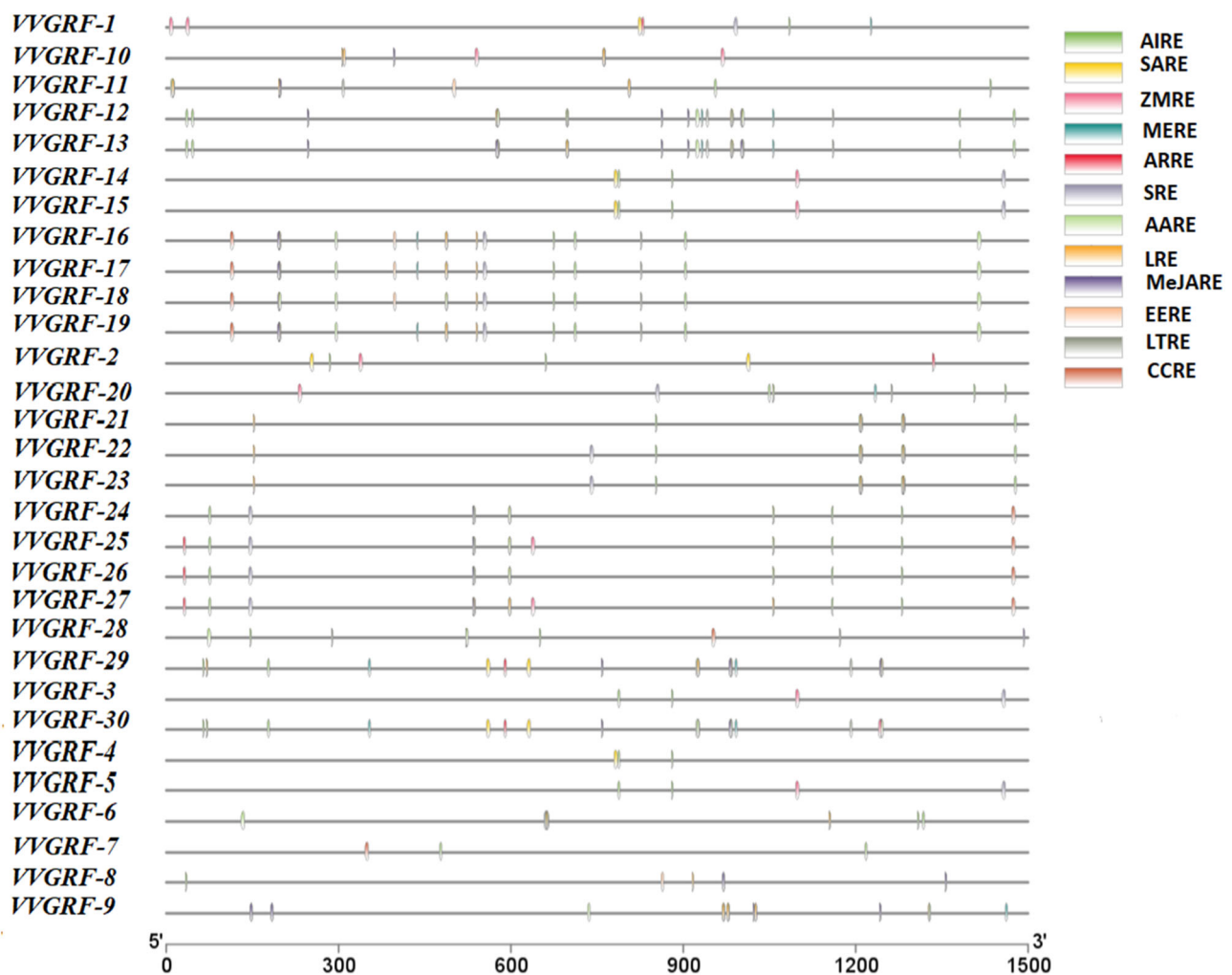


Fig. 4 Cis-regulatory element analysis of *VvGRF*. TFs Gene structure of the *GRF* gene family

Group I (seven *VvGRFs*, four *FvGRFs*, and two *AtGRFs*), 10 *GRF* members in Group II (three *VvGRFs*, three *FvGRFs*, and two *AtGRFs*), five *GRF* genes in Group III (three *VvGRFs*, one *FvGRF*, and one *AtGRF*), six members in Group IV (two *VvGRFs*, two *FvGRFs*, and two *AtGRFs*), nine *GRF* genes (all *VvGRFs*), and six members (all *VvGRFs*) (Fig. 6). Group 1 was the largest group. No species-specific clades were found, and *VvGRFs* were present in any group (Fig. 6). The phylogenetic tree revealed a strong link between *VvGRFs*, *FvGRFs*, and *AtGRFs*, suggesting that *GRF* proteins have been conserved throughout evolution.

Motif distribution of *GRFs* in *Vitis vinifera*

We identified ten conserved motifs (1–10) in *VvGRF* proteins. Conserved motif analysis revealed that the amino acid sequence of the WRC region is more conserved than that QLQ region. The combined block diagram of 10 conserved motifs among the identified *GRF* genes in *Vitis vinifera* is

represented by different colors and *p*-values (Fig. 7A). Similar patterns were observed for most *GRF* members in the same subclade, indicating that they might have similar functions. Most *GRF* proteins comprised nine motifs, including *VvGRF-1* to *VvGRF-4* and *VvGRF-15* to *VvGRF-18*. Sequences of the conserved motifs are also provided. Motif 1, shown in red in the box, is present in all the *VvGRF* genes. The conservation of these motifs across proteins indicated their functional similarity (Fig. 7B, C). Hence, conserved motifs may point to possible functional locations or show that genes are involved in the induction of related downstream functions.

Conserved domain alignment

Two conserved sites in *GRF* proteins are QLQ (Gln, Leu, Gln) and WRC (Trp, Arg, Cys). A transcription coactivator complex is formed when the *GRF* interaction complex (*GIF*) and QLQ domain engage. The *GRF* protein's DNA-binding region, known as the WRC domain, is comprised of

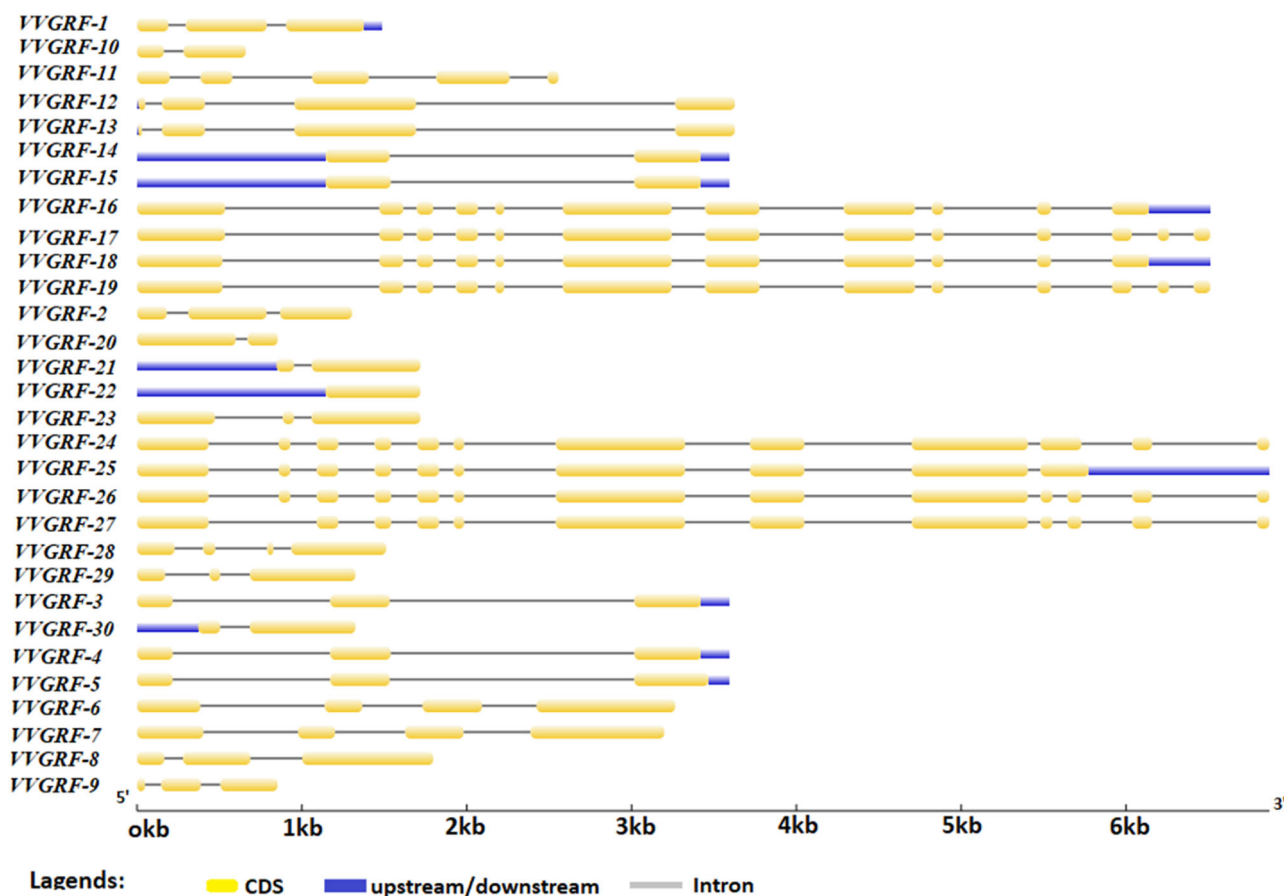


Fig. 5 Gene structure of *VvGRF* genes

a zinc finger structure that binds to DNA and a nuclear localization signal motif. Through domain analysis, various superfamilies were identified, including the cupin-RmIC-like superfamily (represented by purple color), the QLQ superfamily (represented by orange color), the WRC superfamily (represented by yellow color), and the zf-4CXXC-R1 superfamily (shown by blue color) (Fig. 8). All the *VvGRF* genes contain members of the WRC superfamily. The *VvGRF*s gene (4,2,3,19,20,21,22,1) contains the Cupin-RmIC-like superfamily, and gene numbers (4,2,3,19,20,21,22,1) contain the Zf-4CXXC-R1 superfamily. Thus, genes that contain the zf-4CXXC-R1 superfamily also belong to the cupin-RmIC-like superfamily.

Protein-protein interaction network

We found that there may be direct, indirect, or reciprocal interactions between the *GRF* proteins. D7SVF8-VITVI, a zf-GRF domain-containing protein, was used as the input and was found to interact with multiple proteins including F6H9P7-VITVI, SKIV2L2, F616P7-VITVI, E0CSL1-VITVI, E0CUI8-VITVI, F6GUC0-VITVI, F6GV52-VITVI, F6GXX7-VITVI, F6HC46-VITVI, and F6HWC2-VITVI. F6H9P7-VITVI,

SKIV2L2, E0CSL1-VITVI, E0CUI8-VITVI, F6GXX7-VITVI, F6HC46-VITVI, and F6HWC2-VITVI are uncharacterized proteins, while F6GV52-VITVI is an NTP-transf-2 domain-containing protein and F616P7-VITVI is an DSHCT domain containing protein. This PPI network indicated that the interacting proteins had similar functions (Fig. 9).

Expression data analysis of *Vitis vinifera*

The GSE175475 dataset was used for expression analysis of *VvGRF* between the control and yeast-treated samples to identify the role of these genes in the biotic stress conditions. *VvGRF-5* and *VvGRF-6* genes were highly upregulated in the yeast samples, suggesting that these genes are involved in biotic response. The upregulation of these genes indicates their role in the defense mechanism of the *Vitis vinifera* (Fig. 10A). The *VvGRF-10*, *VvGRF-16*, and *VvGRF-23* were found to be highly downregulated in the yeast samples, which indicated that these genes are involved in the growth mechanisms of the plants and suppressed as part of stress response (Fig. 10B). During the biotic stress conditions, the plant mimics the growth mechanism (*VvGRF-10*, *VvGRF-16*, and

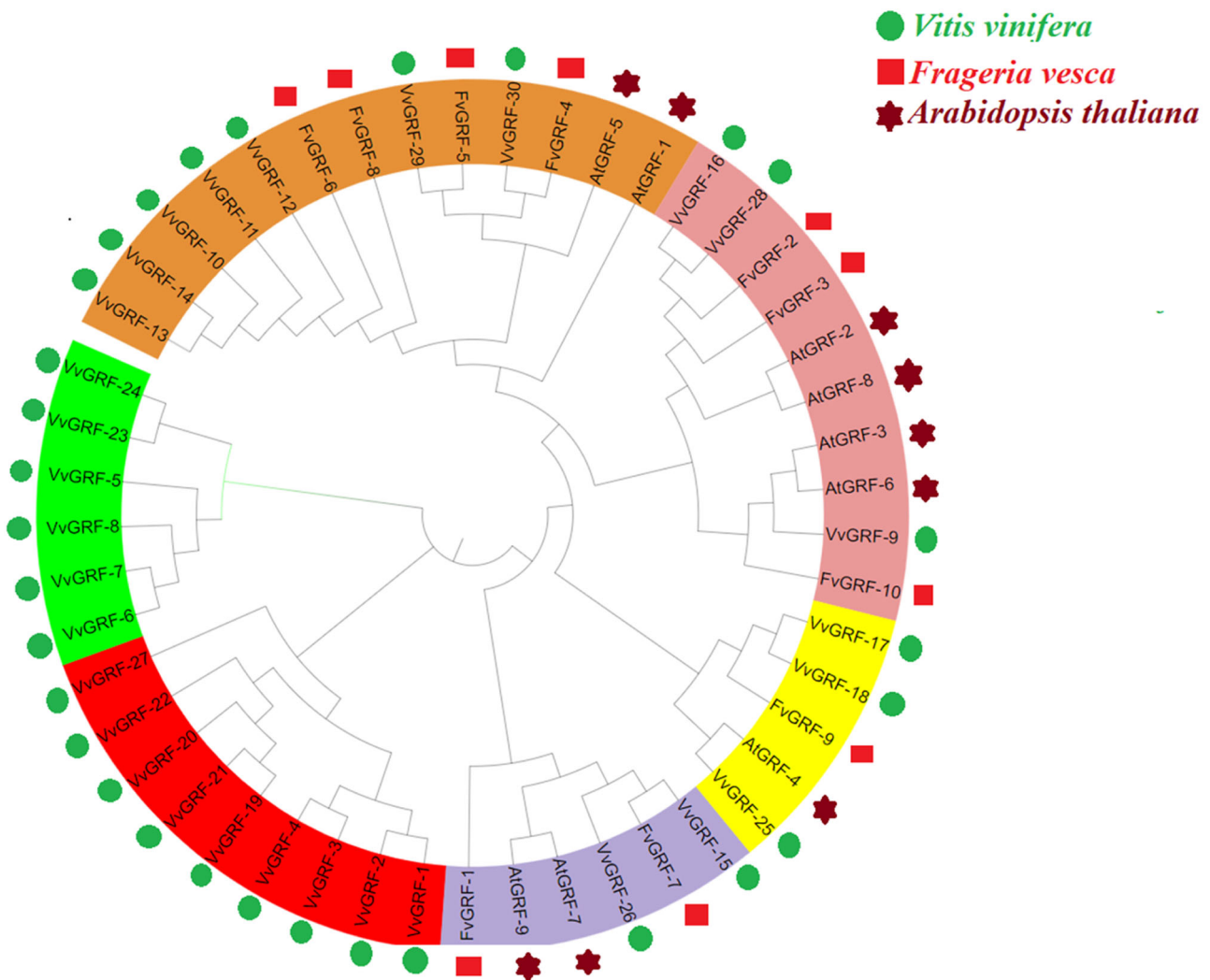


Fig. 6 Phylogenetic tree analysis of *Vitis vinifera*, *Arabidopsis thaliana*, and *Frageria vesca*

VvGRF-23) and enhances the defense mechanisms (*VvGRF-5* and *VvGRF-6*).

Prediction of the miR396 target sites of *VvGRFs*

MicroRNAs are single-stranded, non-coding RNAs of 20–24 nucleotides that regulate gene expression of target genes and inhibit translation or promote mRNA destruction [22]. We examine different miRNA binding sites in 30 *VvGRF* genes, including MiR396a, MiR396b, MiR396c, and MiR396f, which target multiple *VvGRF* genes. MiR396a interacts with *VvGRF-2* to *VvGRF-15*. MiR396b interacts with *VvGRF-1* to *VvGRF-11*, *VvGRF-13*, *VvGRF-14*, and *VvGRF-15*; MiR396c targets *VvGRF-1* to *VvGRF-10*, *VvGRF-12* to *VvGRF-15*, and MiR396f targets *VvGRF-1* to *VvGRF-5*, *VvGRF-7*, *VvGRF-8*, *VvGRF-10* to *VvGRF-13*, and *VvGRF-15* (Fig. 11). Thus, distinct *GRF* genes interact with numerous miRNA target sites. The

microRNAs are specified and target only one gene. Most miRNAs prevent cleavage, whereas others prevent the targeted genes from being translated.

Gene ontology analysis

The GO ontology study was conducted to reveal essential functional characteristics of *VvGRF* genes. Gene ontology (GO) terms were divided into three categories: biological process, cellular component; and molecular function. The significant biological processes include leading strand elongation, DNA repair, base-excision repair, and the metabolic process of nucleic acids. Molecular function terms were enriched in exodeoxyribonuclease, polynucleotide 3-phosphatase, DNA binding, DNA N-glycosylase, and the synthesis of 5-phosphomonoesters. Cellular components include DNA replication and base-excision repair (Fig. 12A).

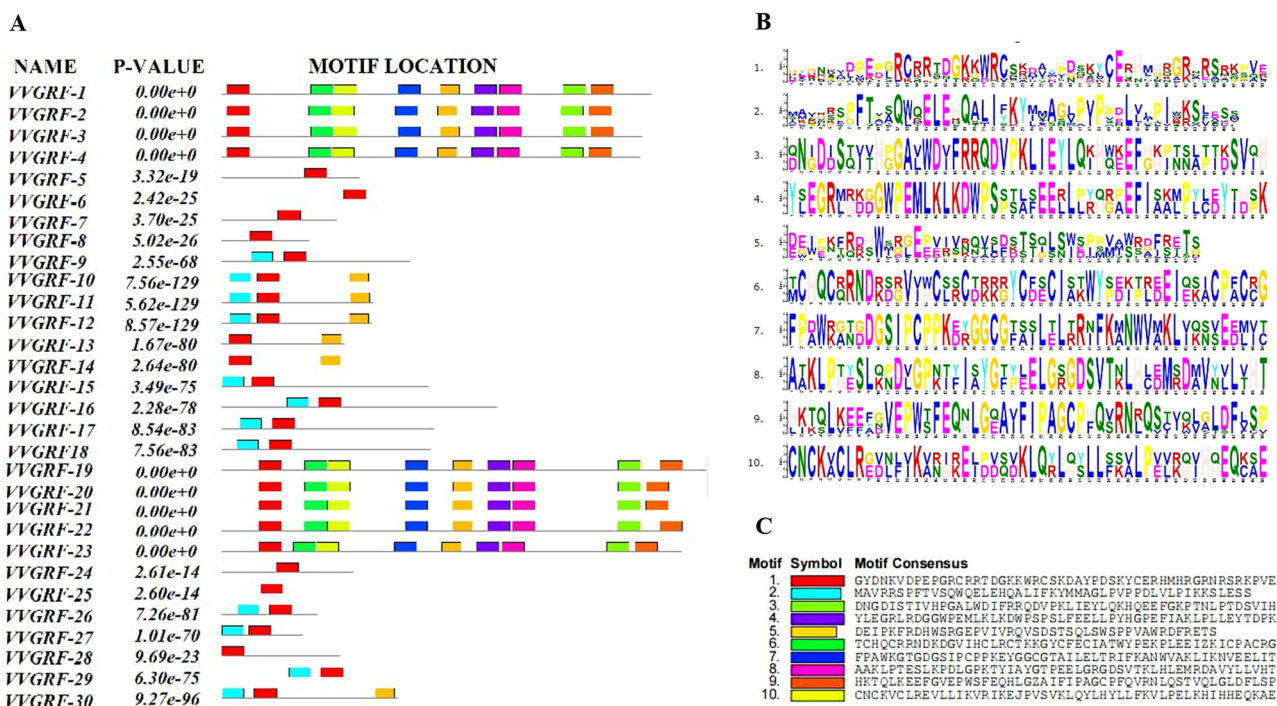
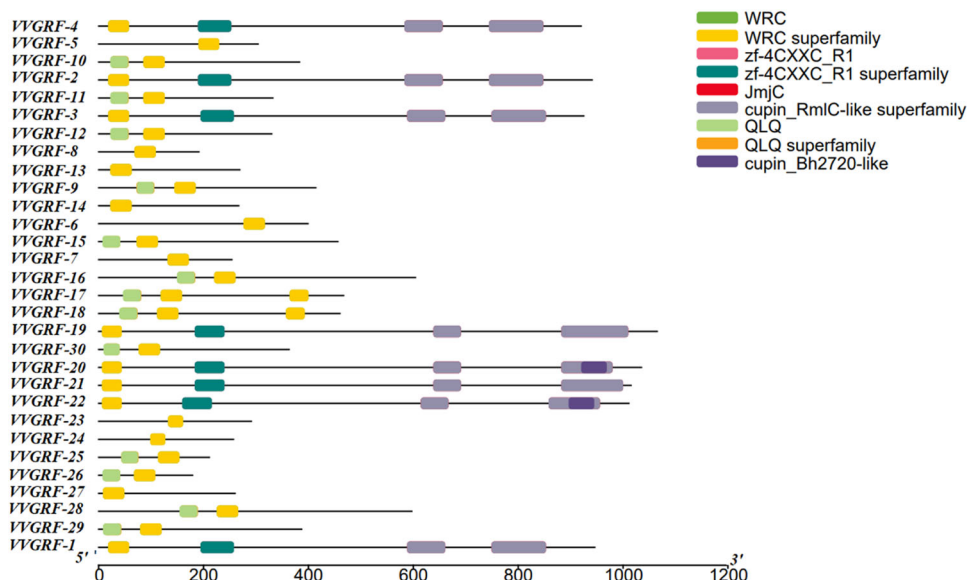


Fig. 7 **A** The 10 motifs of *VvGRF* proteins, represented as rectangles with distinct colors. **B** Graphical representation of conserved motifs **C** Conserved motif sequences

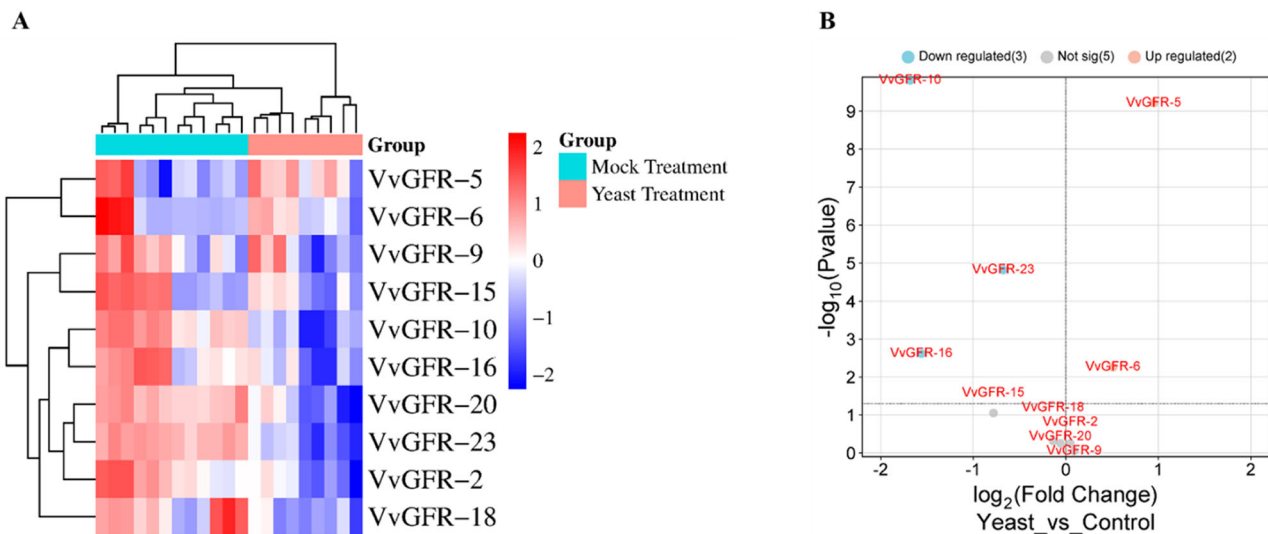
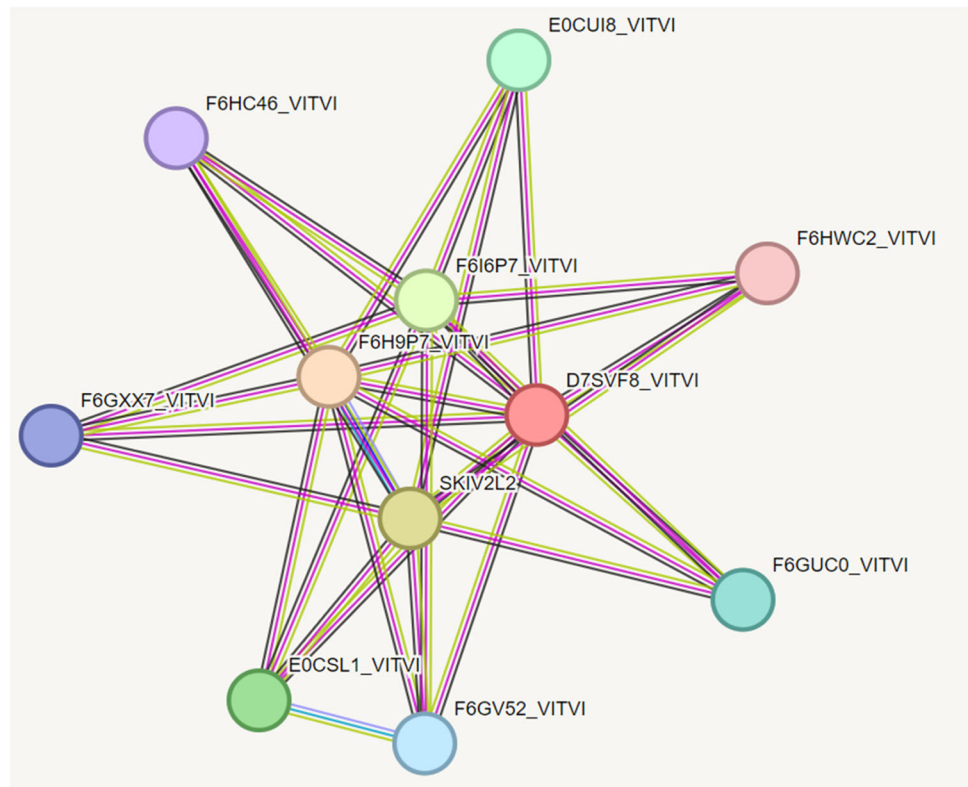
Fig. 8 Domain analysis of *VvGRF* genes



Sub-cellular localization of *VvGRF* genes in different organelles

Although 30 GRF genes were identified in the *Vitis vinifera* genome, there is still no information available about the physiological functions of these genes. We investigated how *VvGRFs* are expressed in the *Vitis vinifera*

nucleus, cytoplasm, mitochondria, chloroplast, endoplasmic reticulum, and other organelles. Localization analysis showed that the *VvGRF* protein is located in various organelles, such as mitochondria, chloroplasts, cytosol, cytoplasm, nucleus, endoplasmic reticulum, and plasma membrane (Fig. 12B). Different expression pattern was observed in *VvGRF*.

Fig. 9 Protein-protein interaction network**Fig. 10** Expression data analysis of *Vitis vinifera*. **A** Heatmap of the expression data. **B** DEGs between control and yeast treatment samples

Discussion

Growth regulatory factors are crucial for the development and growth of plants, particularly for controlling the size of their organs [8]. The *GRF* family contains two functional domains, including the P-P interaction motif QLQ (glutamine, leucine, and glutamine), which is involved in

chromatin remodeling, and the plant-specific WRC (tryptophan, arginine, and cysteine) domain, involved in DNA binding and TF induction to the nucleus. Preserving these motifs in *GRF* genes enables computational characterization and identification of these gene families from emerging genomes [27]. Many researchers have discovered and studied the genetic functions of the *GRF* transcription factor in

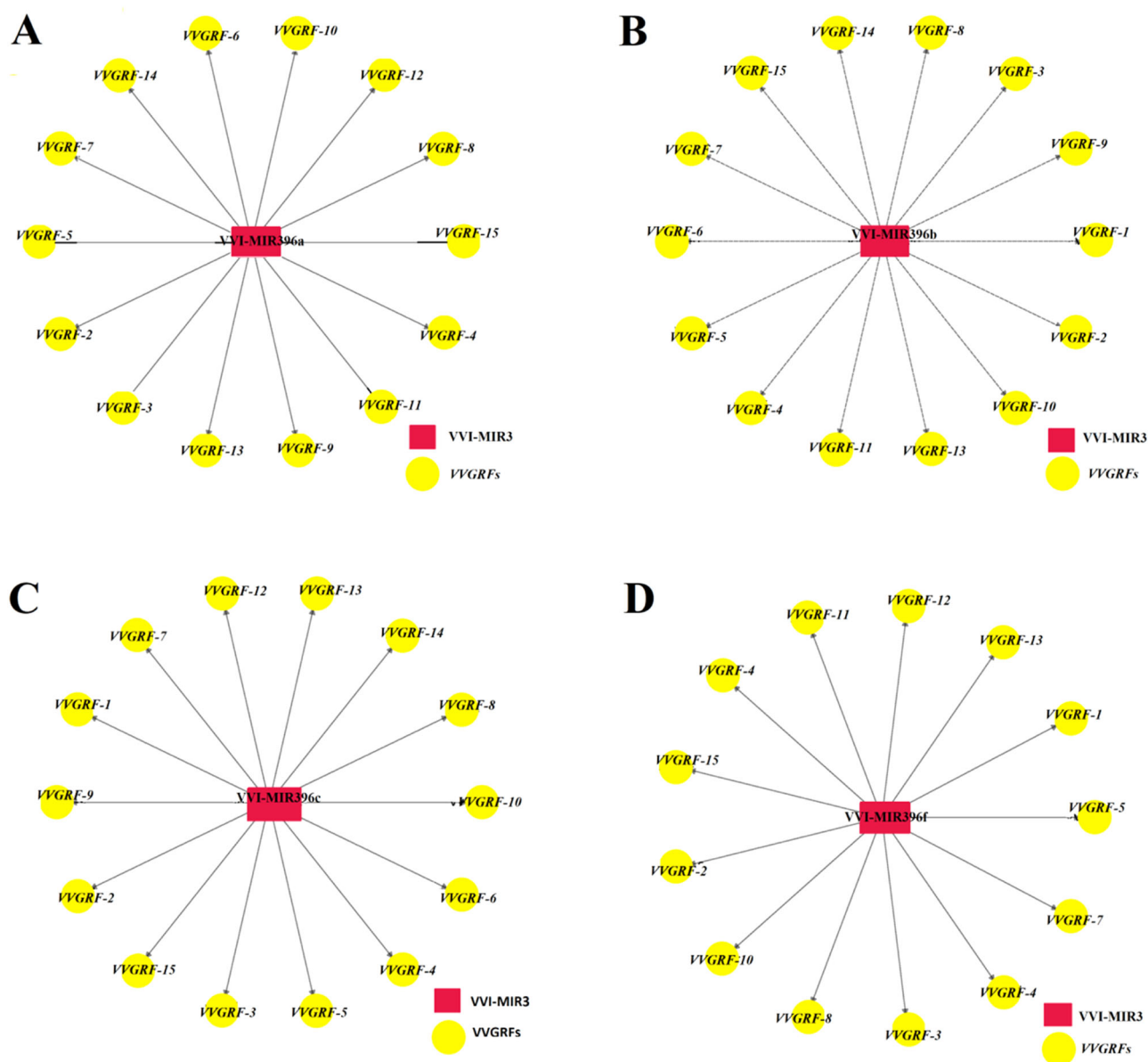


Fig. 11 Different miRNA targets. Red Nodes represent the miRNA, and yellow nodes represent the VvGRF

various plants. The results of these studies indicate that GRF genes are mainly expressed in plant meristems and are crucial for plant development and growth [5].

In this study, we identified 30 GRF genes from *Vitis vinifera*. The number of GRF genes varies among different plants. We identified nine AtGRFs in *Arabidopsis thaliana* [7], twelve OsGRFs in rice [28], fifty-six TaGRFs in wheat [29], seventeen BrGRFs in *Brassica rapa* [30], nine CsGRFs in orange [31], and eighteen PeGRFs in bamboo [32]. The physicochemical properties of the 30 GRF genes in the *Vitis vinifera* genome were analyzed to find their differences in protein clades. All VvGRF proteins were found to express hydrophilic properties, indicated by negative GRAVY values indicating the potential to interact

with water, and also have electrical charges based on pH levels. In phylogenetic analysis of 30 VvGRF TFs with 9 AtGRFs and 10 FvGRFs in *Vitis vinifera* were divided into six groups. According to past studies, the phylogenetic tree of GRF in rice can be categorized into three groups [28]. In contrast, in *Arabidopsis* and *Citrus sinensis* there are five groups [7, 33], in wheat there are seven groups [29], in bamboo, three groups are present [34], and in Chinese cabbage there are six groups [30]. These findings suggest that the GRF transcription factors in monocotyledons differ from those in dicotyledons regarding their evolutionary patterns and traits.

The subcellular localization analysis illustrates that VvGRF proteins were localized among multiple

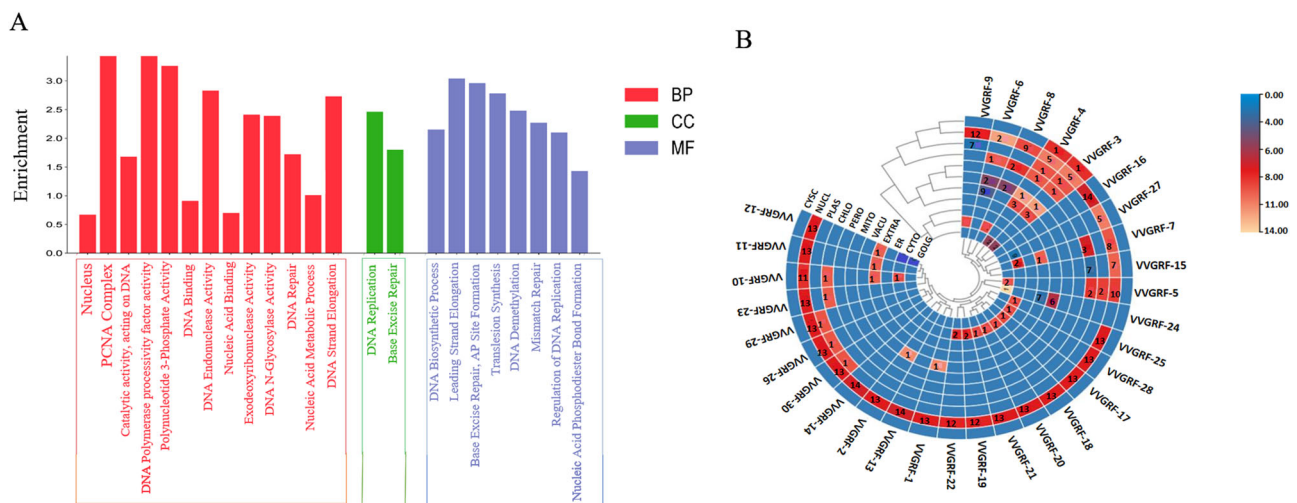


Fig. 12 **A** Gene ontology bar chart of *GRF* genes. **B** Subcellular localization of *GRF* genes. Red indicates the maximum expression ratio, while blue represents the lowest ratio

organelles, such as the mitochondria, chloroplast, cytosol, cytoplasm, nuclei, endoplasmic reticulum, and plasma membrane [20]. Previous studies have shown that the location of exons and introns in gene families is important for evolution [35]. All *VvGRF* genes contained exons and introns. The domain and motif of *GRF* are highly conserved. *Cis*-regulatory elements investigation reveals that the promotor region of *VvGRF* contains 12 CREs, including AIRE, SARE, ZMRE, MERE, ARRE, SRE, AARE, LRE, MeJARE, EERE, LTRE, and CCRE, which play an important role in the development and growth of plants and their responses to hormones, stress, light, and temperature.

Gene ontology analysis of *VvGRF* genes highlights the potential role of *GRF* TFs in grapevine, such as in biological processes, molecular function, and cellular components. Four miRNA targets were identified: MiR396a, MiR396b, MiR396c, and MiR396f, that control gene regulation mechanisms in plants. In transcriptome analysis, expression data of the GSE227706 dataset was analysed and filtered. This extensive analysis of gene expression data revealed that *VvGRF-5* and *VvGRF-6* are upregulated under yeast treatment which indicate its role in the biotic response. During the biotic stress conditions, the plant mimics the growth mechanism (*VvGRF-10*, *VvGRF-16*, and *VvGRF-23*) and enhances the defense mechanisms (*VvGRF-5* and *VvGRF-6*).

The results of this study shed light on the evolutionary history and functional diversity of the plant *GRF* gene family. These findings are an invaluable resource for further research, enabling functional analysis and cloning of these genes. This study's thorough genome-wide identification and characterization will make it possible for future research.

Conclusion

The *GRF* gene family in *Vitis vinifera* was identified and thoroughly analyzed for the first time in this study. In *Vitis vinifera*, 30 *GRF* genes were identified. The 23 *GRF* genes out of 30 are located on 8 of the 20 chromosomes and are divided into six subgroups in phylogenetic analysis. CREs in the promotor region of *VvGRF* genes respond to light, hormones, defense, and growth. Multiple miRNA targets that might be involved in controlling the expression of the *VvGRF* gene have also been identified. Gene structure, conserved domain and motif, subcellular localization, and protein-protein interaction network analysis are performed using different bioinformatic tools and databases. Expression data for *Vitis vinifera* was also analyzed. This computational study reveals the vast genome-wide understanding of grapevine *GRF* genes. These findings encourage more investigation into the roles and regulatory processes of *GRF* transcription factors in the growth and development of *Vitis vinifera*.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1007/s12013-024-01554-9>.

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Author Contributions All author contributes to the paper.

Compliance with ethical standards

Conflict of interest The authors declare no competing interests.

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